

词汇: *space exploration* 太空探索

- 1 For years, astronomers have **hypothesised** about life beyond Earth, especially on the closest planet to Earth – Venus. Venus is often named the Earth's 'evil twin' because of its extremely hot temperature and **greenhouse effect**, which make it toxic and **inhospitable**. It is also similar in size, mass and rocky **composition**.
- 2 Though no **organic molecules** have been discovered on Venus, new findings from the Synthetic Aperture **Radar** (SAR) instrument on board NASA's Magellan spacecraft shows caves **lurking** beneath the surface. Researchers from Nature Communications add that these could offer us new ideas about the evolution of Venus – about how it might have possibly supported life in the past or could again in the future.
- 3 The caves suggest that lava tubes, otherwise known as pyroducts, could be **prevalent**. The search for pyroducts was challenging at first because the planet is **shrouded** in thick clouds of sulphur and carbon dioxide that not only contribute to the **immense** temperature but also block any view of the surface. However, the recent radar images suggest that a big pyroduct lies near Nyx Mons, a large volcano in the **highlands** of Venus's Bell Regio region.
- 4 In a 2026 study called 'Radar-based observation of a lava tube on Venus', it can be understood that, in the field of **geoscience**, volcanoes play an important role in the evolution of a planet. If we think about Earth billions of years ago, volcanoes released gases that helped form our oceans and our atmosphere. Volcanoes keep Earth warm, wet and **habitable**. So, we have evidence that there are many active volcanoes on Venus, but will we ever be able to live there?
- 5 Well, we would require **an abundance of** water, which is greatly **lacking** on Venus. But data gathered by the Venus Express orbiter, launched in 2005 by The European Space Agency, could **hint** toward the planet once having oceans. And though the clouds on Venus rain sulphuric acid, they could still support an **exotic biochemistry** that **thrives** in such conditions. There's also plenty of sunlight to serve as a power source for life.

测验与练习

一、为本文选择总结句。

- a. Evidence tells us there is life on Venus.
- b. There's evidence life may have been possible on Venus.
- c. We will all move to Venus in the future.

二、根据文章内容选出正确的选项。

1. Which is not a reason for why we call Venus Earth's 'twin'?
 - a. They are made of rock.
 - b. They are similar in size.
 - c. They both have toxic atmospheres.
2. Proof of life has recently been discovered in Venus's caves.
 - a. True
 - b. False
 - c. Not given
3. It's difficult to observe Venus's surface because the view is blocked by...
 - a. its clouds.
 - b. its highlands.
 - c. the Sun.
4. Which is not a reason for why volcanoes helped the evolution of Earth?
 - a. Volcanoes decreased the temperature so that life could grow.
 - b. Volcanoes helped form our oceans.
 - c. Volcanoes released the gases that make our atmosphere.
5. We don't need lots of water to survive.
 - a. True
 - b. False
 - c. Not given

三、用方框中的单词完成对本文的总结。

So far, scientists have not discovered anything that is **1)** _____ on Venus. It's no surprise as the planet is completely **2)** _____. This is because of the **3)** _____ heat, toxic gases and sulphuric rain. However, **4)** _____ studies have revealed volcanic activity that could suggest, not only an evolution of the plane, but a specific **5)** _____ that lives there.

biochemistry	geoscience	habitable
hint		inhospitable
immense		organic

（答案见词汇表后）

词汇表

hypothesise	假设，假定
greenhouse effect	温室效应
inhospitable	无法居住的，条件恶劣的
composition	组成，构成
organic	有机的
molecule	分子
radar	雷达
lurk	潜藏
prevalent	普遍存在的
shroud	笼罩，掩盖
immense	极高的
highland	高原
geoscience	地球科学
habitable	可居住的
an abundance of	大量的
lacking	缺乏的，不足的
hint	展现出...的迹象，暗示
exotic	奇特的
biochemistry	生物化学
thrive	茁壮成长

答案

一、为本文选择总结句。

b. There's evidence life may have been possible on Venus.

二、根据文章内容选出正确的选项。

1. c. *It is also similar in **size, mass and rocky composition**.*

2. b. **False.** *Though **no organic molecules have been discovered on Venus**, new findings of caves lurking beneath the surface could offer us new ideas about the evolution of Venus.*

3. a. *The search for pyroducts was challenging at first because the planet is shrouded in **thick clouds of sulphur and carbon dioxide**...*

4. a. *If we think about Earth billions of years ago, volcanoes **released gases that helped form our oceans and our atmosphere**.*

5. b. **False.** *Well, **we would require an abundance of water** which that is greatly lacking on Venus.*

三、用方框中的单词完成对本文的总结。

So far, scientists have not discovered anything that is **organic** on Venus. It's no surprise as the planet is completely **inhospitable**. This is because of the **immense** heat, toxic gases and sulphuric rain. However, **geoscience** studies have revealed volcanic activity that could suggest, not only an evolution of the plane, but a specific **biochemistry** that lives there.